GGG-L-51a
FEBRUARY 15, 1957

SUPERSEDING
Int. Fed. Spec. GGG–L–51a (Navy–Ships)
September 8, 1955, and
Fed. Spec. GGG–L–51
March 1, 1932

FEDERAL SPECIFICATION

LADLE, MELTING, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers ladles generally used by plumbers for transporting and pouring molten metals and occasionally used for melting metals such as lead, solder, and babbitt.

1.1.1 *Federal specification coverage.*—Federal specifications do not cover all types and sizes of the commodity indicated by the title of the specifications, but are intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types.*—Ladles shall be of the following types as specified (see 6.2) :

Type I.—Double lip.
Type II. — Single lip and spout (bottom pouring).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specification and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Military Specification:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standard:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

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Nongovernmental:

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules and Container Specifications.

(Application for copies of the Consolidated Freight Classification Rules and Container Specifications should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations shown herein are descriptive and not restrictive, are included for the convenience of identification, and are not intended to preclude the purchase of ladles which are otherwise in accordance with this specification.

3.2 Material.—The material from which the ladle is constructed shall be of a quality steel, malleable iron, wrought iron, cast iron, or iron alloy possessing high heat retentivity qualities. The ladles shall be capable of resisting thermal shock when subjected to the test specified in 4.3.

3.3 Finish.—All ladles shall have a smooth, even surface, thoroughly cleaned and substantially free from scale, pits, nodules, burrs, cracks or other detrimental defects, with a finish conforming to good commercial practice.

3.4 Design.—The handle of the ladle shall be either integral or attached to the bowl in such a manner to preclude its working loose under service conditions.

3.5 Identification marking. — Each ladle shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.6 Type I, double lip.—Type I ladles shall be provided with two lips on the top rim of the bowl. Each of the lips shall be located at a right angle to the center line of the handle. The ladles shall conform to the dimensions shown in table I for the diameters specified (see 6.2) and shall be similar to figure 1.

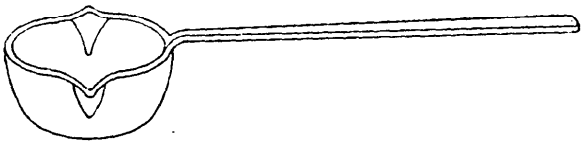


FIGURE 1.—Type I, double-lip ladle.

3.7 Type II, single lip and spout (bottom pouring).—Type II ladles shall have a bottom-pouring spout extending from the top of the rim to the bottom of the bowl opposite the lip. The opening of the spout at the bottom of the bowl shall be of a size to properly allow pouring of metals from the bottom of the bowl. The outside base of the bowl shall

TABLE I.—*Type I, double-lip ladle*

Diameter of bowl ± ¼	Depth of bowl		Thickness of bowl		Length of handle	
	Min.	Max.	Min.	Max.	Min.	Max.
Inches	*Inches*	*Inches*	*Inch*	*Inch*	*Inches*	*Inches*
2½	1	1¼	3/32	⅛	9	11½
3	1⅛	1½	3/32	⅛	11	11½
4	1 1/16	1¾	3/32	5/32	12	15½
5	2	2½	⅛	5/32	18½	19½
6	2 5/16	2⅝	⅛	7/32	20	22½
7	2⅞	3¼	5/16	7/32	24	31

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be bossed or flattened to prevent tilting and the spilling of metals when placed on the floor or other flat surfaces. The outer end of the handle should be provided with an eye or similar means whereby the ladle can be easily tilted. Type II ladles shall be provided with a sliding and revolving type of hand-gripping sleeve whereby the ladle can be supported and easily tilted. The ladles shall conform to the dimensions shown in table II for the diameter specified (see 6.2), and shall be similar to figure 2.

TABLE II.—*Type II, single lip and spout (bottom pouring)*

Diameter of bowl ± ¼	Depth of bowl ± ¼	Length of handle ± ½	Weight (approx.)
Inches	*Inches*	*Inches*	*Pounds*
3	2¼	12¼	2½
4	2	16	3½
5	2½	24	5½
6	3	29	9
7	4	29	10½

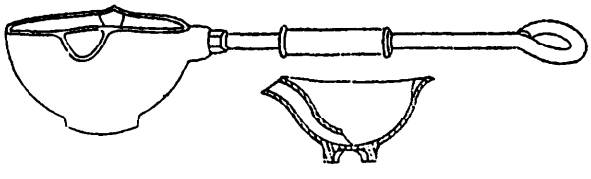


FIGURE 2.—Type II, single lip and spout (bottom pouring).

3.8 **Workmanship.**—Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice. The ladles shall be free from any imperfections that may affect their durability or serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.* — All ladles of the same type and size presented at one time shall be considered a lot for the purpose of acceptance inspection and test.

4.1.2 *Sampling for inspection and test.*— From each lot a random sample of ladles shall be taken for inspection in accordance with Military Standard MIL–STD–105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300, and inspection level I for lots of 301 and over. The acceptance quality level (AQL) shall be 1.5 percent defective.

4.2 Visual and dimensional inspection. — Each of the sample ladles taken in accordance with 4.1.2 shall be visually and dimensionally inspected by the inspector to verify compliance with this specification. Surfaces shall be inspected for visible cracks, blowholes, sand inclusions, and dark colored, chilled spots. Any ladle in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective ladles in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Test. — The sample ladles shall be heated to 700°F., plus or minus 20°F. and quenched in water at approximately 70°F., and shall be subjected to a light tap on the side of the bowl to obtain a resonant ring. A flat nonresonant ring indicates an invisible crack and shall be cause for rejection, and if the number of defective ladles in any sample exceeds the acceptance number for that sample the lot shall be rejected.

4.4 Rejected lots.—Rejected lots may be resubmitted for inspection and test, provided the contractor has removed or repaired all nonconforming ladles. The inspector shall again take and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

(For definitions and levels see 6.1.)

5.1 Preservation and packaging.

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5.1.1 *Level A.*—Level A cleaning, preservation and packaging shall be in accordance with level A in Military Specification MIL–P–15424 (see 6.2).

5.1.2 *Level C.*—Level C preservation and packaging shall afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may be the supplier's commercial practice when such meets the requirements (see 6.2).

5.2 Packing.

5.2.1 *Levels A and B.*—Levels A and B packing shall be in accordance with levels A and B respectively in Military Specification MIL–P–15424 (see 6.2).

5.2.2 *Level C.*—Level C packing shall insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Consolidated Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation (see 6.2).

5.3 Marking of shipments.

5.3.1 *Military.*—Marking of shipments to military agencies shall be in accordance with Military Specification MIL–P–15424.

5.3.2 *Nonmilitary.*—Unless otherwise specified for civilian governmental agencies, shipping containers shall be marked with the name of the commodity, the type, the size, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order (see 6.2).

6. NOTES

6.1 Nonmilitary purchases. — Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type of ladle (see 1.2).
(*c*) Diameter of ladle (see tables I and II).
(*d*) Level of preservation, packaging and packing required (see section 5).
(*e*) Marking of shipments for civilian governmental agencies, if different (see 5.3.2).

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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Navy—Sh A MC Or Y
Air Force.

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